

DSA 8020 Statistical Methods II

Project I

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Due Date: February 20 (Thursday), 11:59pm ET

Instructions

For each problem, your answer should include the following elements:

- A **descriptive summary** of the data using summary statistics and at least one plot/graph or table. Give your description in paragraph form.
- An appropriate **inferential analysis** to address the question. This may be a confidence interval, hypothesis test, or both. You can use the α value/confidence level of your choice. If you perform a hypothesis test, state the hypotheses, test statistic, p-value, decision, and conclusion. If you create a confidence interval, state the confidence level and provide an interpretation of the interval.
- A thoughtful **conclusion** that summarizes your findings. You should also **discuss any limitations or complications of your analysis**. This should include the following:
 1. An assessment of whether the modeling assumptions are reasonable.
 2. Discussion of any unusual features of the data. For example, do outliers/high leverage points affect your results?

Problem 1

Perform a detailed regression analysis using the `gala` data set. Use `Endemics` as the response variable, excluding `Species` from this analysis, to address the following questions:

1. Provide a **descriptive summary** of the data set.
2. Is at least one of the predictors useful in predicting the response?
3. Perform **model selection** to determine whether all the predictors help to explain the response.
4. Perform a **Box-Cox transformation**¹ to determine if there is a need to transform the response.
5. Compare a **generalized additive model** fit with the linear regression fit and explain why you preferred one model over the other.
6. How well does the chosen model fit the data? Quantify the fit numerically. Additionally, check the model assumptions and summarize your findings.
7. Predict the response value when all the chosen predictors take their median value. Quantify the prediction uncertainty.
8. Provide a thoughtful conclusion that summarize your findings.

¹You will need to add a small positive constant to ensure the response is positive

Problem 2

Perform a detailed analysis using the **Salaries** data set to comment on the salary differences between male and female faculty members.

1. Provide a **descriptive summary** of the data set.
2. Comment on whether there is a need to include interactions between these categorical predictors. Select your model and justify your choice.
3. Perform a **log transformation** on response to determine if there is any improvement in terms of model fit.
4. How well does the chosen model fit the data? Quantify the fit numerically. Additionally, check the model assumptions and summarize your findings.
5. Write a thoughtful conclusion to comment on the salary differences between male and female faculty.